NING Siyuan

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EDUCATION

Master of Science in Physics **2023.08-2024.07**

National University of Singapore

Relevant Coursework: PC4236 Computational Condensed Matter Physics, PC5215 Numerical Recipes with Applications, PC5252 Bayesian Statistics and Machine Learning, PC5251 Applied Machine Learning, PC5253 Complex System Analysis and Modelling, PC5212 Physics of Nanostructures, PC4253 Thin Film Physics and Technology.

Bachelor of Science in Physics, minor in Artificial Intelligence **2019.09-2023.06**

Shandong University

Relevant Coursework: Material Simulation, Computational Physics, Group Theory, Solid State Physics, Film Physics and Techniques; **(minor)** Machine Learning, Neural Network and Deep Learning, Natural Language Processing.

RESEARCH EXPERIENCE

MSc Physics Project supervised by *Prof. QUEK Su Ying,* National University of Singapore **2023.08-2024.07**

Worked on First-principles studies of quantum defect candidates in 2D WS2

- Developed a workflow to investigate 54 substitutional defects in monolayer WS_2 by determining their ground-state structures, defect stability, spin configurations, and defect level positions.
- Identified three spin-triplet defects, and examined their optimized local geometries as well as electronic properties.
- Expedited setup and post-processing of calculations using bash scripts and VASPKIT tookit.
- Wrote a 60-page project thesis and present the results to a panel of examiners [PDF].

Research Assistant under *Prof. Zhang Peng*, Shandong University **2020.06-2022.11**

Worked on Computational Analysis of Hydrogen Bond Vibrations of Ice III in the Far-Infrared Band

- Calculated vibrational spectra of Ice III and its hydrogen-ordered counterpart ice IX, validated by experimental data.
- Analyzed vibration normal modes of hydrogen bonds to explain the origin of far-IR vibrational bands.
- Assessed structural deformations under high pressure in hydrogen bond lengths and included angles.
- Authored first-authored paper published in *Crystal* [PDF].

RELEVANT PROJECTS

Machine learning in Materials Informatics **2024.07**

VASPKIT Workshop

- Employed physics-informed featurization (Matminer) and Interpretable Machine learning techniques (SISSO, SHAP).

Sunspot Number Forecasting: Bayesian Neural Networks (BNNs) via MCMC **[HTML]** **2024.05**

Final Project for *PC5251 Applied Machine Learning*, National University of Singapore

- Built a 10x5x3 neutral network, where parameters are initialized as distributions and updated via Langevin MCMC.
- Obtained high-accuracy predictions with confidence intervals and conducted Gelman-Rubin convergence diagnostic.
- Developed a highly rated Jupyter notebook, recognized for its well-written and pedagogically accessible content.

Deep Learning the Phase Transition: A CNN Approach to the 2D Ising Model **[HTML]** **2023.10**

Research project in *PC5212 Numerical Recipes*, National University of Singapore

- Used the Monte Carlo method to obtain representative sets of spin configurations for a bunch of temperatures.
- Trained convolutional neural networks via PyTorch to classify configurations and predict the critical temperature.

Finite Element Analysis and Optimization of Permanent Magnet Motor **2023.03-2023.06**

Capstone project in minor degree, Shandong University

- Established motor models using ANSYS Maxwell, defining solution domains, boundary conditions, and mesh subdivisions for basic motor performance simulation.
- Implemented multi-layer band refinement for calculate air-gap flux density and slot torque, followed by a multivariable parameter sweep of relevant structural parameters.
- Proposed structural optimization strategies aligned with output power metrics.

Numerical Simulation on Quantum Size Effects in Nanomaterials **2022.03-2022.11**

Research Project with *Prof. Guan Chengbo*, Shandong university

RESEARCH SKILLS

Computational Modeling Codes/Packages:

- DFT codes: VASP, CASTEP, SIESTA
- Automation & Processing: Vaspkit, Pymatgen
- Model Builder and Visualizer: Material Studio, VESTA
- Finite Element Analysis: ANSYS Workbench

Machine Learning Techniques:

- Frameworks: Scikit-learn, PyTorch, XGBoost, SiSSO
- Ensemble methods: Bagging, Boosting, Stacking
- Explainable tools: SHAP

Programming Language: Python (Numpy, Pandas), Shell (bash), MATLAB

Data Processing: Excel, Origin

VOLUNTEER & TEACHING EXPERIENCE

Volunteer Coordinator | Students' Union, School of Space Science and Physics, Shandong University **2020-2021**

Led and coordinated volunteer activities at nursing homes and schools for children with autism in Weihai.

Teaching Assistant | Teaching Office of General Mathematics, Shandong University **2020-2021**

Assisted in *Calculus* courses by preparing course materials, conducting tutoring, and grading.

GRANTS & AWARDS

Second-class Academic scholarship, Shandong University	2021-2023
Excellent Social Practice Research Report, Shandong University	2021
Volunteer Service Award, Shandong University	2020

THESIS PUBLICATION

Ning, S.-Y.; Cao, J.-W.; Liu, X.-Y.; Wu, H.-J.; Yuan, X.-Q.; Dong, X.-T.; Li, Y.-N.; Jiang, Y.; Zhang, P. *Computational Analysis of Hydrogen Bond Vibrations of Ice III in the Far-Infrared Band.* Crystals 2022, 12, 910. https://doi.org/10.3390/cryst12070910